

# **Module 2**

# **Management Accounting**

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## **Answers**

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The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

**SECTION B – WRITTEN QUESTIONS (Total: 80 marks)****Answer 1(a)**

Budgeted overhead absorbed for PD1 and PD2

|                                     | <u>PD1</u><br>HK\$ | <u>PD2</u><br>HK\$ | <u>SD1</u><br>HK\$ | <u>SD2</u><br>HK\$ | <u>Total</u><br>HK\$ |
|-------------------------------------|--------------------|--------------------|--------------------|--------------------|----------------------|
| Allocated and apportioned overheads | 500,000            | 250,000            | 120,000            | 90,000             | 960,000              |
| Apportion SD1                       | 36,000             | 60,000             | (120,000)          | <u>24,000</u>      |                      |
|                                     |                    |                    |                    | 114,000            |                      |
| Apportion SD2                       | 74,100             | 39,900             | -                  | (114,000)          |                      |
| Budgeted overhead                   | 610,100            | 349,900            | -                  | -                  | 960,000              |

**Answer 1(b)**

PD1 is machine intensive in that it incurs a high level of machine hours (400,000 machine hours), and incurs a low level of direct labour hours (10,000 direct labour hours) in the coming year. PD1 should use machine hours as its overhead absorption base.

On the other hand, PD2 is labour intensive in that it incurs a high level of direct labour hours (150,000 direct labour hours) and incurs a low level of machine hours (20,000 machine hours) in the coming year. PD2 should use direct labour hours as its overhead absorption base.

**Answer 1(c)**

Budgeted overhead absorption rate:

PD1: HK\$610,100 ÷ 400,000 = HK\$1.53 per machine hour

PD2: HK\$349,900 ÷ 150,000 = HK\$2.33 per labour hour

**Answer 1(d)**

Quotation for the job:

|                                                    | HK\$             |
|----------------------------------------------------|------------------|
| Direct materials                                   | 200,000          |
| Direct labour:                                     |                  |
| PD1 (HK\$60 x 1,000)                               | 60,000           |
| PD2 (HK\$55 x 10,000)                              | 550,000          |
| Production overheads:                              |                  |
| PD1 (HK\$1.53 x 20,000)                            | 30,600           |
| PD2 (HK\$2.33 x 10,000)                            | 23,300           |
|                                                    | <hr/>            |
| Production costs                                   | 863,900          |
| Gross profit margin of 20% (HK\$863,900 x 20 / 80) | 215,975          |
| Selling price                                      | <u>1,079,875</u> |

**Answer 1(e)**

Under or over absorption of overheads

|                           | <u>PD1</u>                          | <u>PD2</u>                          |
|---------------------------|-------------------------------------|-------------------------------------|
| Overhead absorbed         | HK\$1.53 x 380,000<br>= HK\$581,400 | HK\$2.33 x 180,000<br>= HK\$419,400 |
| Overhead incurred         | HK\$560,000                         | HK\$475,000                         |
| Over / (Under) absorption | HK\$21,400                          | (HK\$55,600)                        |

**Answer 1(f)**

The main reason for many companies using pre-determined, but not actual, overhead absorption rate is that actual overheads and actual consumption of absorption base are not known until the end of an accounting period. It is not realistic to wait until the end of the accounting period to determine the overheads to be allocated to the products.

Products are made throughout the year and the use of a pre-determined overhead absorption rate enables overheads to be absorbed immediately after production to facilitate the determination of the total product cost and setting the selling price.

Moreover, when a pre-determined overhead absorption rate is used, the production overheads absorbed in each output unit tend to be averaged out in the year, while the actual production overheads per unit may fluctuate from time to time in the year.

**Answer 2(a)**

Production budget (in units):

|                                      | Gaea<br>Units      |
|--------------------------------------|--------------------|
| Budgeted sales                       | 8,200              |
| Budgeted inventory increase          | 800                |
|                                      | <hr/> 9,000        |
| 10% loss on output (9,000 x 10 / 90) | 1,000              |
| Production budget                    | <hr/> <hr/> 10,000 |

**Answer 2(b)**

Direct material purchase budget:

|                                                | Agul<br>Kg          | Belki<br>Kg       |
|------------------------------------------------|---------------------|-------------------|
| Direct material usage for 10,000 units of Gaea | 200,000             | 100,000           |
| Increase (decrease) in inventory               | 3,000               | (1,000)           |
| Direct material purchases                      | <hr/> 203,000       | <hr/> 99,000      |
| Price per kg                                   | HK\$15              | HK\$10            |
| Purchase costs                                 | <hr/> HK\$3,045,000 | <hr/> HK\$990,000 |

**Answer 2(c)**

Direct labour budget:

| Product             | Hours            | HK\$           |
|---------------------|------------------|----------------|
| Gaea (10,000 x 1.5) | 15,000 at HK\$60 | <u>900,000</u> |

**Answer 2(d)**

Pre-determined factory overhead absorption rate:

Budgeted factory overheads / Budgeted direct labour hours

= HK\$600,000 / 15,000 = HK\$40 per direct labour hour

### **Answer 2(e)**

Budgeted factory cost of goods produced:

|                                     | Gaea<br>HK\$ |
|-------------------------------------|--------------|
| Direct materials:                   |              |
| Agul (200,000 x HK\$15)             | 3,000,000    |
| Belki (100,000 x HK\$10)            | 1,000,000    |
|                                     | <hr/>        |
|                                     | 4,000,000    |
| Direct labour                       | 900,000      |
| Factory overheads                   | 600,000      |
|                                     | <hr/>        |
| Factory cost of goods produced      | 5,500,000    |
| Good output units                   | 9,000 units  |
| Unit factory cost of goods produced | 611          |

### **Answer 2(f)**

Management accounting uses accounting and financial management to protect and increase the value of an enterprise.

Management accounting is used to identify and generate information, interpret and use that information to develop business strategies, plan and control the operations of a business, make decisions, ensure the efficient use of resources, improve processes, protect assets and ensure effective corporate governance and controls.

Strategic management accounting is a form of management accounting in which emphasis is placed on information which relates to factors external to the firm, as well as non-financial information and internally generated information.

### **Answer 3(a)**

Make or buy decision:

|                                    | HK\$  |
|------------------------------------|-------|
| Relevant costs per unit:           |       |
| Direct materials                   | 32    |
| Direct labour                      | 65    |
| Variable production overheads      | 20    |
| Avoidable fixed costs              |       |
| [HK\$(420,000 – 300,000) ÷ 10,000] | 12    |
|                                    | <hr/> |
|                                    | 129   |
| Unit cost of purchase              | <hr/> |
|                                    | 140   |

QIL should make the required components in-house as the unit cost of purchase is higher than the relevant costs of production.

**Answer 3(b)**

New decision:

|                                                           |            |
|-----------------------------------------------------------|------------|
|                                                           | HK\$       |
| Original relevant costs per unit                          | 129        |
| Opportunity cost of leasing out<br>(HK\$150,000 ÷ 10,000) | 15         |
| Revised relevant costs per unit                           | <u>144</u> |
| Unit cost of purchase                                     | <u>140</u> |

With the leasing income, QIL should buy the required components from the outside supplier as the unit cost of purchase is lower than the relevant costs of production.

**Answer 3(c)**

The non-financial factors to be considered include:

- (i) Whether the outside supplier can deliver the components on time. Late deliveries could affect the company's production schedule and delivery dates for the final products.
- (ii) Whether the quality of components bought from outside the supplier is equal to, or better than, the quality of components made internally. A low quality of the components will negatively affect the quality of the final products and will thus affect the company's sales and reputation in the long run.
- (iii) If the components are made outside, redundancies may result which could affect employee morale and cause labour problems.

**Answer 4(a)**

|                                                    | Zhuhai<br>Division<br>HK\$             | Chongqing<br>Division<br>HK\$             |
|----------------------------------------------------|----------------------------------------|-------------------------------------------|
| Investment<br>(Total assets – current liabilities) | 3,600,000 – 2,100,000<br>= 1,500,000   | 2,200,000 – 900,000<br>= 1,300,000        |
| Return on investment ("ROI")                       | 250,000 / 1,500,000 x<br>100% = 16.67% | 180,000 / 1,300,000 x<br>100%<br>= 13.85% |
| Required return (15% x investment)                 | 225,000                                | 195,000                                   |
| Residual income ("RI")                             | 250,000 – 225,000<br>= 25,000          | 180,000 – 195,000<br>= (15,000)           |

**Answer 4(b)**

An acceptable investment should have an ROI that exceeds the firm's required rate of return ("RRR"), which is the minimum return a firm's management will accept for an investment. Equally, an acceptable investment has a positive RI. A positive RI indicates a return greater than a firm's RRR.

The Zhuhai Division has a better performance. Its ROI is higher than the RRR and it has a positive RI, which is acceptable to the management.

The Chongqing Division has an ROI lower than the RRR and a negative RI. The management of the company may consider shutting down the division and putting the investment thereon to other profitable use or to explore new geographical markets.

\* \* \* END OF EXAMINATION PAPER \* \* \*